

THE SELF-HOSTED AI LEARNING PLATFORM ↘

LEARN THE FUTURE WAY.

An AI tutor fine-tuned on real Cambridge past papers — grounded in **official mark schemes**, running entirely on your own server. Zero third-party AI APIs. **Live and deployed today.**



Your data never leaves your server

LoRA fine-tune of Qwen2.5-0.5B runs in-process — no OpenAI, no vendor cloud.



Answers you can trust

Retrieval over 68k+ aligned questions returns the official mark scheme, attributed.



Snap & Solve vision

Photograph a diagram — the moondream2 VLM sees it, OCR fallback included.

BY THE NUMBERS

68k+

ALIGNED PAST-PAPER Q/A

\$0

PER-QUESTION AI COST

3

ROLE-BASED PORTALS

100%

SELF-HOSTED STACK

WHAT MAKES IT DIFFERENT ↴

NOT A GPT WRAPPER.



Self-hosted, fine-tuned tutor

A LoRA fine-tune of Qwen2.5-0.5B on aligned Cambridge papers. Most edtech resells a vendor API — we own the model.



Mark-scheme-grounded answers

Retrieval returns the official Cambridge mark scheme, attributed — not a hallucinated guess.



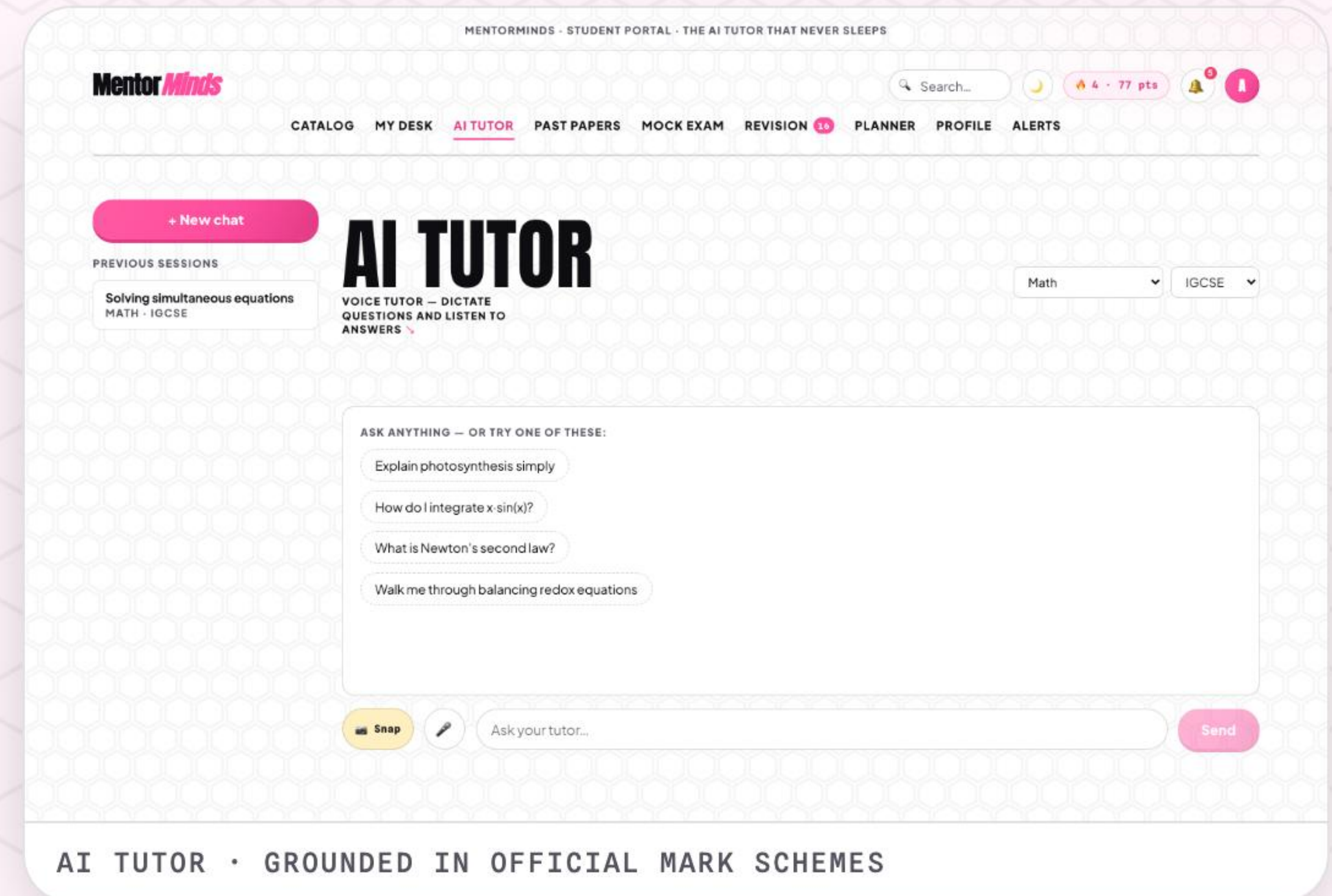
Snap & Solve VLM

moondream2 reads photographed diagrams & handwriting; clean MCQs become auto-marked questions.



Adaptive learning loop

Weak-topic detection, spaced repetition, and a dropout-risk model close the loop.



Try it: **AI Tutor** → ask a past-paper question → attributed mark scheme

mentorminds.duckdns.org

ONE STACK, FULLY SELF-HOSTED ↴

ENGINEERED, NOT ASSEMBLED.

3 Angular SPAs
STUDENT · STUDIO · ADMIN

▼ HTTPS · Caddy (auto-TLS) · Docker Compose · VPS

Django + DRF API
AUTH · COURSES · PLANNER

Channels WS
LIVE CHAT

Celery + Redis
JOBS · CACHE · QUOTAS

PostgreSQL
PRIMARY + REPLICA

▼ X-API-Key

FastAPI ml-service (HF Space)
QWEN2.5-0.5B + LORA · RAG OVER 68K+ Q/A

moondream2 VLM
/V1/VISION/ASK · OCR FALLBACK

OpenCV
PROCTORING · OMR · OCR

Live on VPS + HF Space

CI green – tests + builds

Dynamic-first – live feature flags

Sovereign – no third-party AI APIs

\$0 marginal inference

A working product, not a mockup – every feature has a deterministic fallback

github.com/Arnobrizwan/mentormind

REAL STUDENTS, REAL OUTCOMES ↘

EXAM-GRADE TUTORING, FOR EVERYONE.



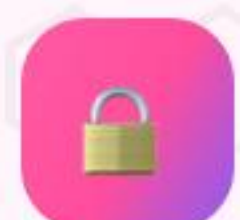
SDG 4 — Quality Education

Mark-scheme-grade tutoring for students who can't afford private tutors.



\$0 per-question cost

A 0.5B model on cheap CPU — schools in emerging markets can run it sustainably.



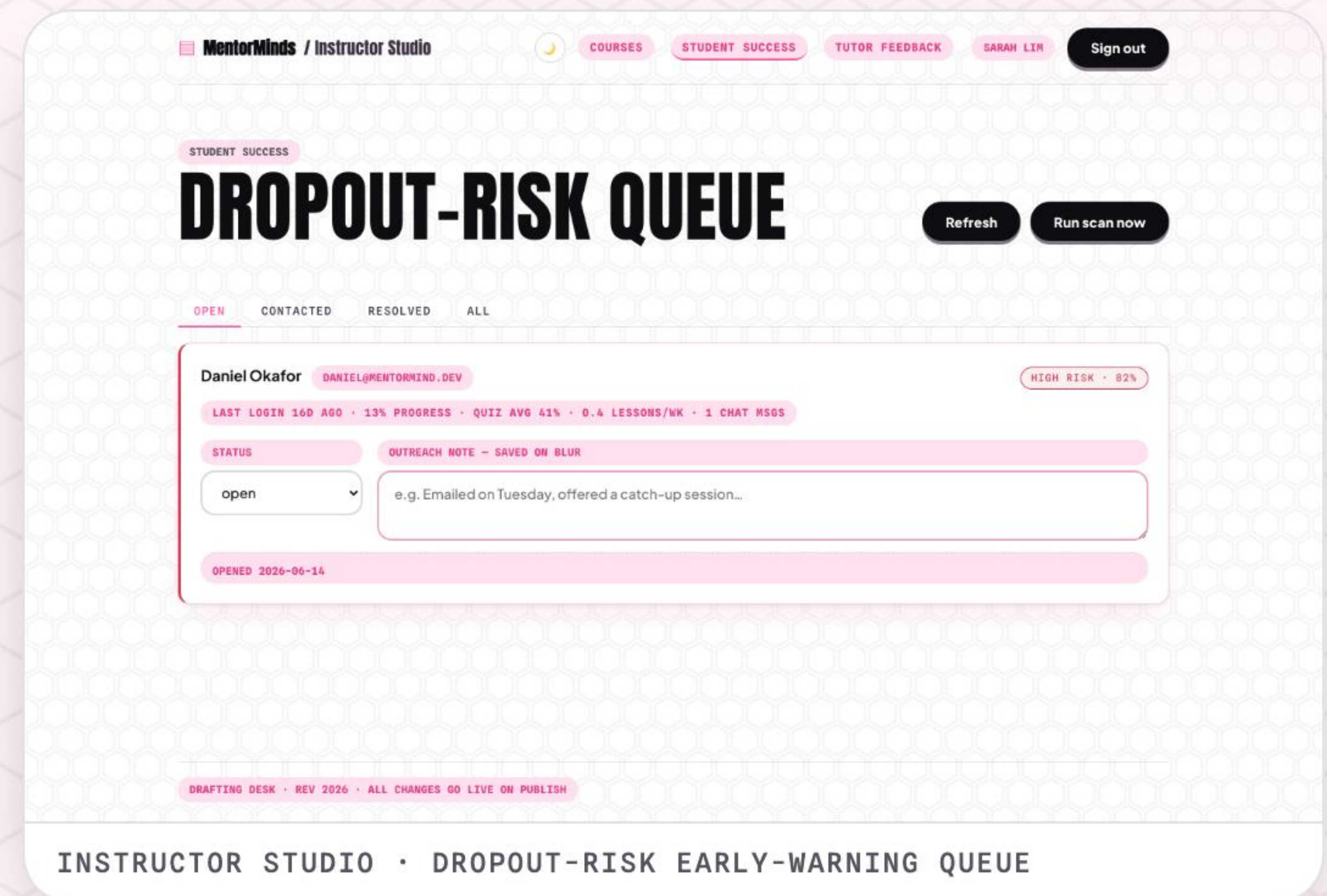
Data sovereignty

Schools self-host; student data never leaves their server.



Dropout-risk early warning

An ML model surfaces at-risk students so instructors intervene before they quit.



ONE BOLD DESIGN SYSTEM

THREE PORTALS,
ONE STACK



Cohesive EduNova design

One bold, accessible visual language across all three portals.



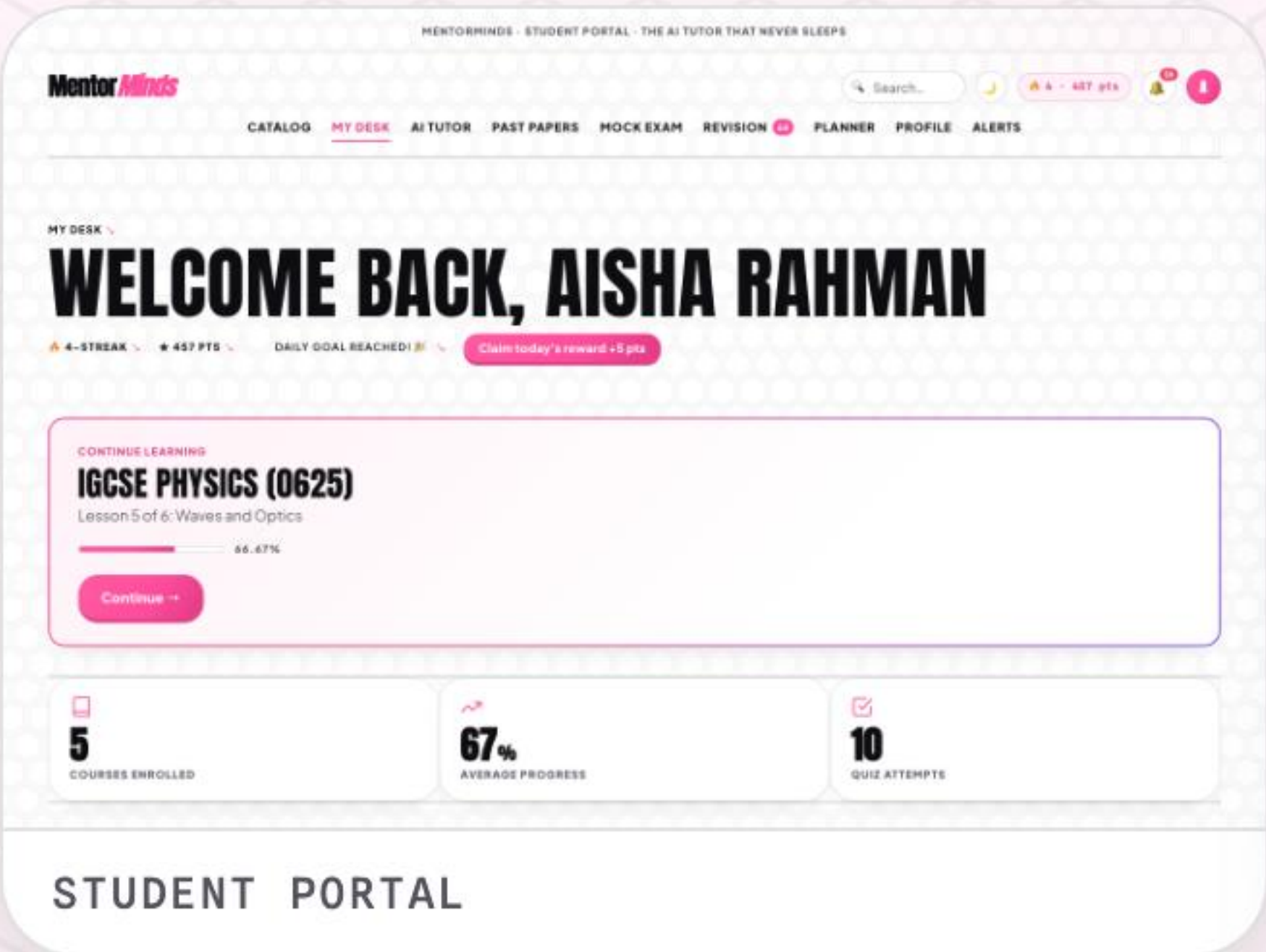
Judges click the real thing

Not slideware — a live, seeded demo with login. URLs on the last slide.

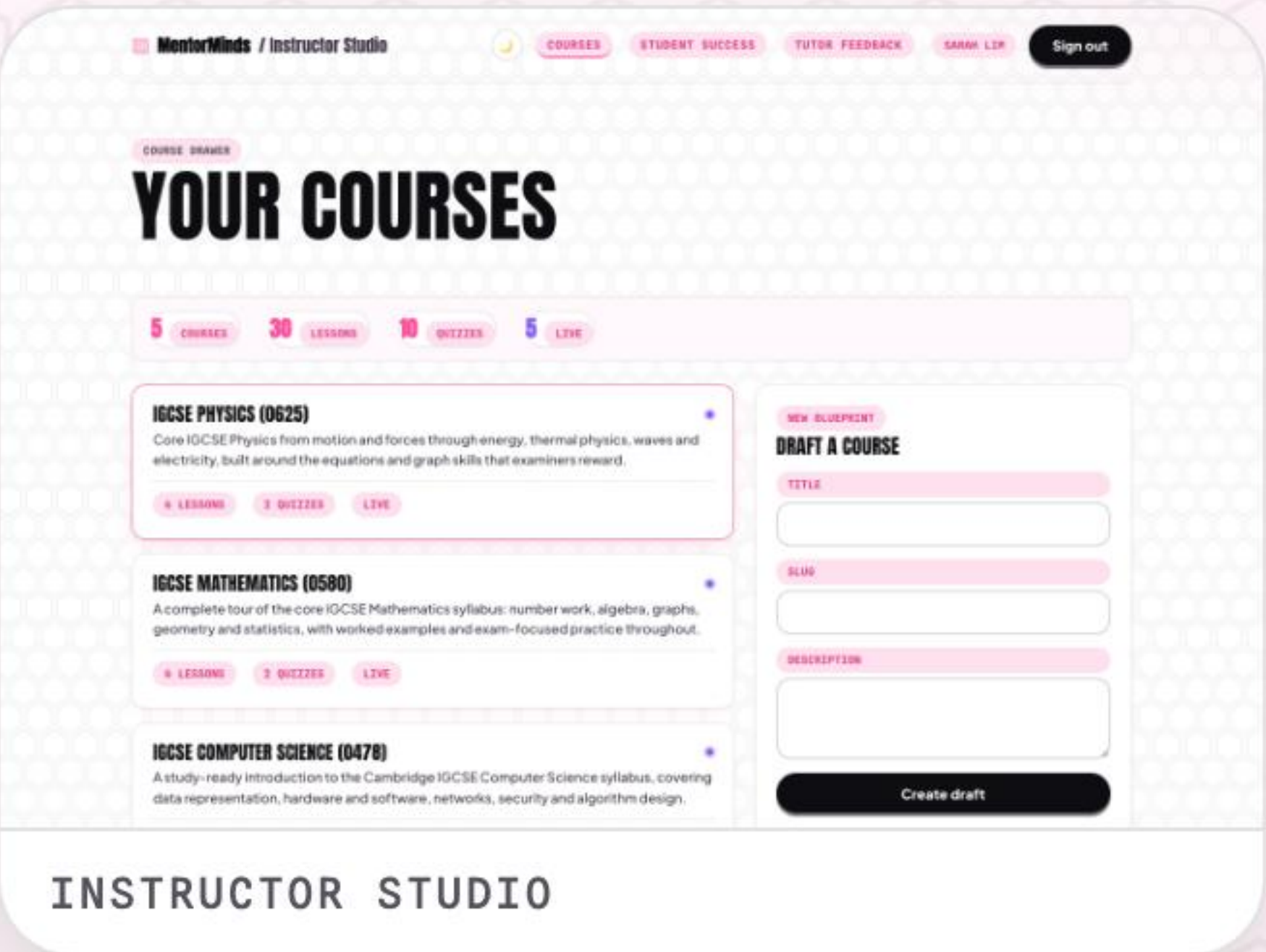


Responsive + Android

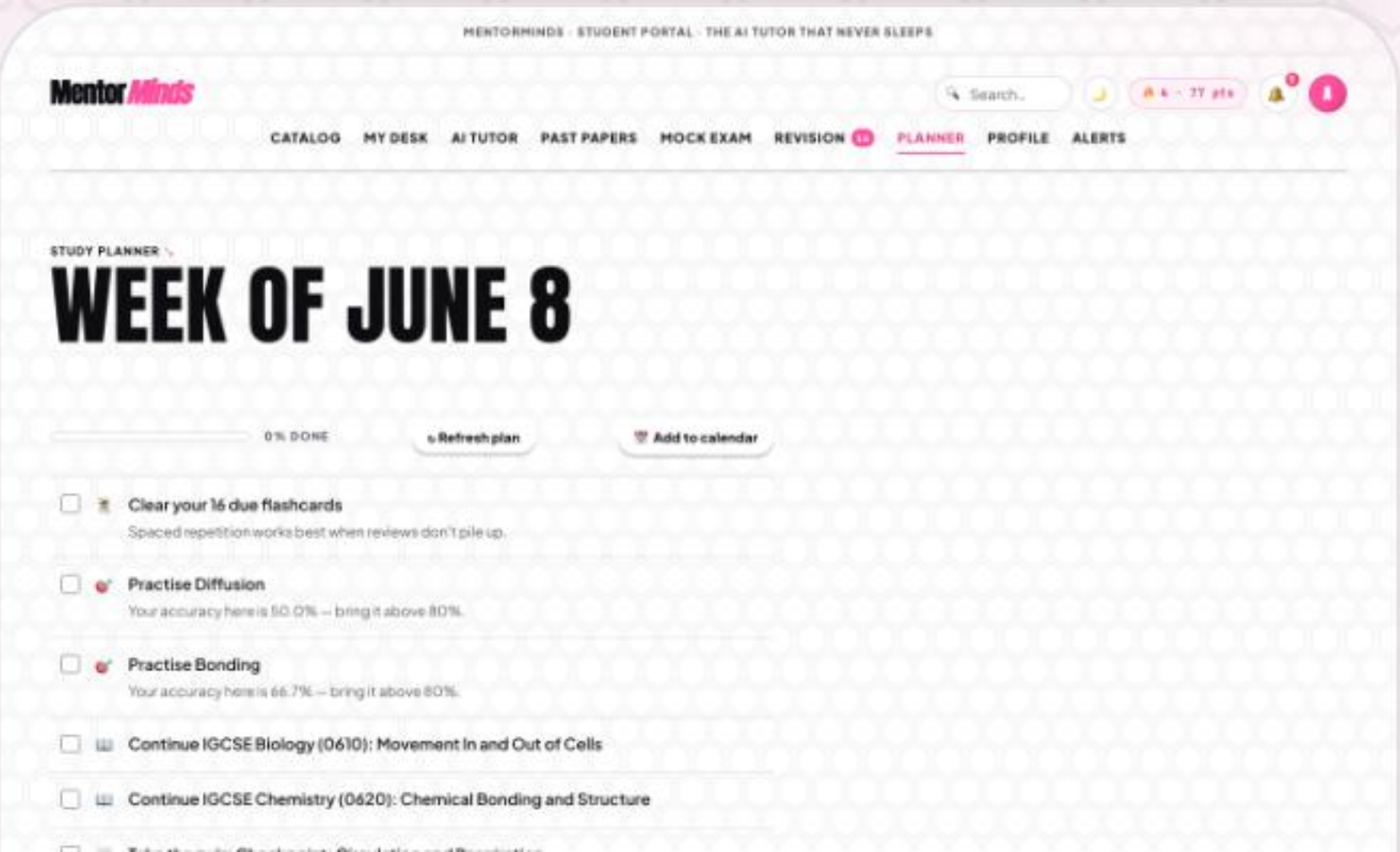
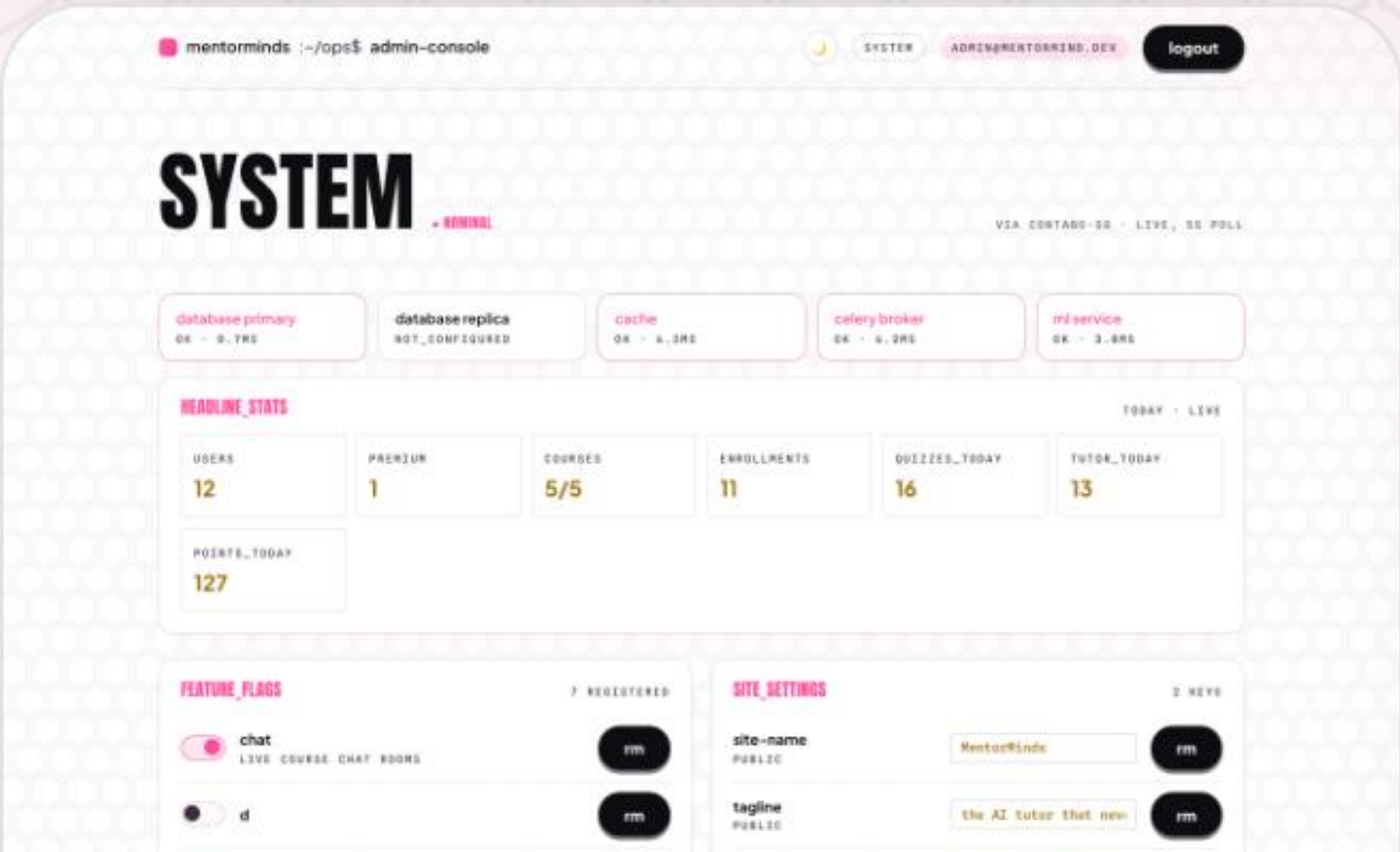
A Capacitor wrapper ships the student portal as a native app.



STUDENT PORTAL



INSTRUCTOR STUDIO



A MODEL THAT PAYS FOR ITSELF ↴

BUILT TO SCALE IN BANGLADESH.



Target market

O/A-Level & IGCSE students and coaching centres across Bangladesh.



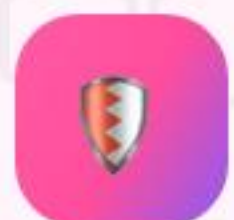
Dual revenue

Per-school self-host licence (B2B) + premium student subscription (B2C, already live), paid via bKash/Nagad.



High-margin & scalable

\$0 marginal AI cost + one multi-tenant stack = margin grows per school.



Defensible moat

Proprietary 68k+ aligned-paper corpus + a fine-tuned model rivals can't API-call.

UNIT ECONOMICS

\$0

COST PER AI ANSWER

B2B+B2C

REVENUE STREAMS

1

STACK, MANY SCHOOLS

68k+

CORPUS MOAT

HOW THE BUSINESS FITS TOGETHER ↘

BUSINESS MODEL CANVAS

 KEY PARTNERS

- BD schools & O/A-Level coaching centres — reach students
- bKash · Nagad · SSLCommerz — payment rails
- Hugging Face — hosts the AI inference
- Open-model makers: Qwen2.5, moondream2

 KEY ACTIVITIES

- LoRA fine-tuning on past papers
- Curating the 68k+ aligned corpus
- Platform dev & self-host support

 KEY RESOURCES

- Proprietary 68k+ Q/A corpus (moat)
- Fine-tuned Qwen2.5 + LoRA model
- 3-portal full-stack platform

 VALUE PROPOSITIONS

- **Mark-scheme-grounded** AI tutoring — answers you can trust
- **\$0 per-question** AI cost — self-hosted & affordable
- **Data sovereignty** — runs on the school's own server
- All-in-one: tutor · past papers · mocks · revision · planner · proctoring

 CUSTOMER RELATIONSHIPS

- Self-serve, gamified student portal
- Instructor dashboards + dropout-risk alerts
- Per-school onboarding

 CHANNELS

- Direct B2B sales to schools
- Self-host licence + web SaaS
- In-app premium · Android app

 CUSTOMER SEGMENTS

- O / A-Level & IGCSE students in Bangladesh (B2C)
- English-medium & coaching-centre learners
- Schools & coaching centres (B2B)

 COST STRUCTURE

- Low fixed infra (VPS + HF Space) ▪ Occasional model training
- Engineering / product ▪ ~\$0 marginal inference cost

 REVENUE STREAMS

- B2B: per-school self-host licence / SaaS tiers ▪ B2C: premium student subscription (live)
- Add-on: institutional analytics ▪ High margin — AI cost ≈ \$0

DON'T TAKE OUR WORD FOR IT ↘

**CHECK EVERY
CLAIM, LIVE.**



mentorminds.duckdns.org

Student portal · login student@mentormind.dev /12345678



/studio & /console

Instructor studio & admin ops console, same domain.



ml-service API

arnob666666-mentormind-ml.hf.space/docs — every ML endpoint, live.



Source + CI

github.com/Arnobrizwan/mentormind — green builds & tests.

